

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-a41f563cc502700d1.jsonl`  
- SHA-256: `2d9ecfd305c6a60bb66c5f995b3df65e7b5ace520d2501a82c3597068d63cec0`  
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```

Transcript

1. user (2026-06-12T14:39:59.545Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).

Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [medium] H6-GREEN requires an unprompted complaint, but Q6 is the prompt ? and the sheet can't record ordering

Same structural bug as H5: PMF_FIELD_SIGNALS.md line 47 grades H6-GREEN only when they complain about multi-court 'before/without prompting', AMBER when they merely 'agree when asked'. Q6 (flyer line 43, sheet line 43) is itself a leading question ('does that make video useless today?'). A GREEN can only come from spontaneous mentions during Q1/Q2 ? but the sheet gives the owner-transcriber no way to know whether the quote in the Q6 box came before or after the question was asked.

Proposed fix: Add a checkbox under Q6 on the sheet: 'They brought up multi-court chaos on their own, before I asked: YES / NO.' Optionally soften the Q6 wording in the flyer to be less leading: 'When several courts are in the same frame, what happens to that video?'

2. user (2026-06-12T14:40:05.648Z)

docs/pmf-field-signals

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md

M docs/NEXT_STEPS.md

?? arial.ttf

?? artifacts/

?? docs/FIELD_VISIT_ANSWER_SHEET.md

?? docs/PMF_FIELD_SIGNALS.md

?? scratch/HIGH_RISK_REMEDIATION_PLAN.md

?? scratch/PR1_BODY.md

```
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

3. user (2026-06-12T14:40:11.631Z)

```
1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9      [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12     first visit** ? what those answers *mean* and what totals trigger which decision.
13     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14     15 ambiguous conversations as whatever we hoped to hear.
15
16     ---
17
18     ## 1. The hypotheses under test (each maps to script questions)
19
20     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21     |---|---|---|---|
22     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
28     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31     hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32     roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
33     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37     Score each hypothesis on the answer sheet **from what they said and showed, not
38     what you inferred**. When between two grades, take the lower one.
39
40     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41     |---|---|---|---|
42     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
44     | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
45     | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it,
satisfied, sticky |
```

46 | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free" |

47 | **H6 multi-court** | complains about it in their own words before/without prompting | agrees when asked | multi-court footage already works for them |

48 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

49

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral* signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

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55 **Disconfirming-evidence quota:** every visit sheet must record at least one verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

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59

60 **## 3. Pre-registered decision rules (n = 10?15 academies)**

61

62 Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

63

64

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera pilot at the ??3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

69 | **PIVOT A ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

70 | **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C ? payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

72 | **FALSIFIED ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

75

76

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update

79

80 ``PMF_MARKET_RESEARCH.md`` §4 with field-verified rows (today's table is web-sourced).

81

82

83 **## 4. Mechanics**

84

85 - One [`FIELD_VISIT_ANSWER_SHEET.md``](`FIELD_VISIT_ANSWER_SHEET.md`) per academy, same day as the visit, in the associate's words + verbatim quotes.

86

87 - The owner (or Claude session) transcribes each sheet into one row of the rollout table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.

88

89

90 - Demo asset if a conversation earns it: the real compiled reel (`output/MahadevpuraSingles_highlights.mp4``) on a phone ? *show, never promise* (flyer ground rule 3 still applies; the associate doesn't carry the demo, the owner does on follow-ups).

91

92

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```

95     ### Rollup table (one row per completed visit)
96
97     | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98     |---|---|---|---|---|---|---|---|---|---|---|---|
99     | 1 | | | | | | | | | | | |
100
101     ### Tally ? decision read-out (recompute per 5 visits)
102
103     | Metric | Count | Threshold in play |
104     |---|---|---|
105     | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106     | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107     | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108     | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109     | H1 any recording | /n | <3 ? FALSIFIED leg |
110     | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112     ## 5. Guardrails that bind the field work
113
114     - **Phase 1 records nothing** ? notes only (flyer rule 7).
115     - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116       players; **sessions with minors are never recorded for our library** (training-pool
117       guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118     - Answer sheets contain personal contact details ? they stay between the associate
119       and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:40:12.607Z)

```

1     # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3     > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4     > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5     > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7     ## Visit basics
8
9     | | |
10    |---|---|
11    | Date / time | |
12    | Academy name | |
13    | Area (neighbourhood) | |
14    | # courts in the hall | |
15    | Spoke to (name + role) | |
16    | Contact (phone/WhatsApp) ? *only if offered* | |
17    | How long the conversation ran | |
18
19    ## The seven questions ? their answers (verbatim where possible)
20
21    **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23    >
24
25    **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27    >
28
29    **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31    >
32
33    **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34    What's good, what's missing, what does it cost them?**
35

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36      >
37
38      **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39      THEY say first (?)? What was their face doing?**
40
41      >
42
43      **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45      >
46
47      **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48      to call)**
49
50      >
51
52      ## Required: the most negative thing they said (verbatim)
53
54      >
55
56      ## Best quotes (anything vivid, in their words)
57
58      >
59
60      ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score
61      lower)
62      | Hypothesis | Score |
63      |---|---|
64      | H1 recording behavior | GREEN / AMBER / RED |
65      | H2 dead footage | GREEN / AMBER / RED |
66      | H3 audience | GREEN / AMBER / RED |
67      | H4 whitespace | GREEN / AMBER / RED |
68      | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69      | H6 multi-court pain | GREEN / AMBER / RED |
70      | H7 Phase-2 door | GREEN / AMBER / RED |
71
72      ## Competitor block (only if they named one)
73
74      | | |
75      |---|---|
76      | Product | |
77      | What it does well (their words) | |
78      | What it misses (their words) | |
79      | What they pay | |
80
81      ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
82
83      ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
84
85      ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
86      really decides)
87
88      >

```

5. user (2026-06-12T14:40:13.023Z)

```

1      # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3      **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
4      academies are active)
5      **Location:** Bengaluru ? in-person visits to badminton academies
6      **Compensation:** ? ___ per completed academy visit + travel reimbursement *(discussed
7      individually)*

```

```

6      **Reports to:** Avi (phone/WhatsApp: ____ )
7
8      ---
9
10     ## The gig in one paragraph
11
12     We're building an AI tool that takes ordinary practice/match video from a badminton
academy and automatically finds
13     every rally, builds highlight reels, and generates match reports. Before building
further, we need honest, on-the-ground
14     answers from real academies about how they record, what they'd use, and what they'd pay.
**Your job: visit 10?15 good
15     academies, have a friendly 15?20 minute conversation with the owner/head coach using the
script below, and write down
16     what they actually say.** No selling. No tech knowledge needed. If a conversation goes
well, there's a Phase-2 follow-up
17     (coordinating a paid video-recording session) with additional compensation.
18
19     ---
20
21     ## Phase 1 ? The conversation (what you actually do)
22
23     **Who to talk to:** the academy owner or head coach ? the person who decides what the
academy spends money on.
24     If you get a junior coach, be friendly, gather what you can, and ask when the owner is
available.
25
26     **Opening (keep it this honest):**
27     > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton
academies ? it takes normal practice
28     > video and automatically cuts it into rallies and highlights. He asked me to talk to a
few serious academies to
29     > understand how you work with video today. Could I ask you a few questions? Takes 15
minutes, and we're not selling
30     > anything today."
31
32     **The questions (ask in this order; the starred ones matter most):**
33
34     1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired
person?
35     2. ? **Do you have recorded footage sitting around that nobody has time to watch?**
(multi-court hall recordings,
36     tournament days, CCTV) ? roughly how much?
37     3. Who would watch rally clips or highlights if they existed ? coaches for training?
players? parents?
38     4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory,
BadPro+, just YouTube/phone.)
39     What do you like / what's missing in it?
40     5. ? **If you could hand over a full session's video and get back every rally cut out +
a highlight reel + a simple
41     report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per
match?** (Let THEM say a number
42     first. Note their face, not just the words.)
43     6. Your hall has multiple courts running at once, right? Does that make video less
useful today?
44     7. Would you let us record one session here (we pay a small fee for the time) to show
you what the tool produces on
45     YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*
46
47     **After each visit, fill ONE answer sheet ? the template is
[FIELD_VISIT_ANSWER_SHEET.md](FIELD_VISIT_ANSWER_SHEET.md)
48     (print one per visit):** academy name, location, # courts, person spoken to + role +
contact, answers 1?7 (their
49     words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a

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50     Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51     [ `PMF_FIELD_SIGNALS.md` ](PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just
capture honestly.)*
52
53     ---
54
55     ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57     1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58         figure out ? what would feel fair to you?" Never quote a price yourself.
59     2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60         what happened to that file?" gets the truth. Prefer the second kind of question.
61     3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62         (a paid recording session + free sample output) and only as "would you be open to
it".
63     4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64         than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65     5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66         doesn't, and what they pay. Spend extra time here.
67     6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68         and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69     7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70         the academy or players unless the owner offers.
71     8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72     9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73         don't share the answer sheets with anyone except Avi.
74
75     ---
76
77     ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79     You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80         one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81         record 1?2 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82         written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83         our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84         the tool).
85
86     ---
87
88     ## Interested?
89
90     Message Avi at ____ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91         walkthrough) so you're never improvising.
92

```

6. user (2026-06-12T14:42:28.733Z)

```
1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
22     **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23     depending on vendor); (b) a structural shift to **amateur self-recording** on action
24     cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25     segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26     106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27     2024, +45.8% YoY).
28
29     The central tension is the **static-camera constraint**: the tool works best on fixed,
30     full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31     moat (most competitors target team sports / broadcast) and a filter on the addressable
32     user ? the customer must already set a camera on a tripod behind the court. That
33     behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34     the median casual player.
35
36     **Recommended beachhead (strongest single):** **badminton coaching academies and
37     serious club/competitive players in India** ? a large, rapidly formalizing academy
38     ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39     willingness to pay for coaching, an underserved tooling gap (incumbents are
40     team-sports/broadcast), and a direct-but-immature competitor field. The product is
41     *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
42     players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43     **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45     **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46     and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47     software-only BYO-camera model undercuts the hardware players; realistic anchors
48     **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50     **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51     coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52     variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53     China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55     ---
56
57     ## 2. Market sizing & trends
58
59     ### 2.1 Sports analytics / sports-video market
60     - **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61       Insights); other vendors USD 5.6?6.0B. **(2025)*
62     - **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63       ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
```


64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
65 - **Video analytics is the leading segment** within sports analytics (Grand View /
66 Fortune BI) ? the directly relevant sub-market.
67 - **Interpretation (estimate):** the *amateur/club auto-highlight* slice is a small but
68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The
69 product targets the long tail the big numbers under-serve.
70
71 ### 2.2 Action-camera / amateur recording adoption
72 - **2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
73 - **Sports is the largest end-use segment** of action cameras (GMIInsights, 2024).
74 - **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are
75 the de-facto amateur rig.
76 - **Implication:** the capture-side behavior the product depends on is already
77 widespread and growing ? *for the segments that record* ? validating "BYO-camera,
78 software-only."
79
80 ### 2.3 Participation & popularity by sport and geography
81
82 | Sport | Participation signal | Source (year) |
83 |---|---|---|
84 | **Badminton** | 220M+ play regularly; 160+ countries | BWF est. via BadmintonBites
(2023) |
85 | Badminton ? China | ~80M registered, up to ~100M (largest base) | industry
compilations (2023) |
86 | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry
compilations (2023) |
87 | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |
88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |
89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |
90 | **Tennis** | **106M players (2024)** , +25.6% over 5 yrs (87M in 2021) | ITF Global
Tennis Report (2021, 2024) |
91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |
92 | **Pickleball ? US** | **19.8M players (2024)** , +45.8% YoY, +311% over 3 yrs; ~6.2M
core | SFIA / Pickleheads / The Dink (2024) |
93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads
(2024) |
94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly
per PPA claim *(low confidence)* | various (2024?2025) |
95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |
96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders) | Cricket
Rise / Nielsen (2024?2025) |
97
98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
99 (Market.us/Custom Market Insights, 2024?2025).
100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.
101
102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
104 is huge as *fandom* but its amateur self-recording + static-full-court fit is weak
105 (large field, not a fixed-frame court). For *this product's static-camera fit*,
106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
107 item, not a beachhead.
108
109 ---
110
111 ## 3. Customer segment evaluation
112
113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
| B2C/B2B | Verdict |
114 |---|---|---|---|---|
115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
most won't set a tripod | B2C | Volume play, weak economics; use as funnel |
116 | **Coaches & coaching academies** | India academy boom; thousands across Asia |
High ? already sell coaching; USD 300?1,500/yr plausible | Reachable via

federations/distributors/social | ****Strong**** ? already film fixed-angle | B2B/B2B2C | ****Best fit**** ? buyer, recorder, payer aligned |

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary; longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor** ? panning = failure mode** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 ## 4. Competitive & pricing benchmarks

131

Product	Category	Model	Price	Notes (year)
--- --- --- --- ---				
Veo (Cam 3)	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; \$115M raised (2025)
Trace	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro)	Camera leased free with sub (2025)
Pixellot (Air)	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
Spiideo	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
Hudl	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
SwingVision	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr** ; free 2 hrs/mo	Closest analog; 200k+ monthly users; \$10M+ raised (2025)
BadPro+	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
	Niche pickleball/badminton apps	Various	Mixed	Varies Fragmented; few true auto-highlight tools (2025)

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses ***pay-per-use*** ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.

152

153 ---

154

155 ## 5. Recommended beachhead

156

157 Ranked by ****WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.****

158

159

160 ### #1 (strongest) ? Badminton coaching academies & serious players, ****India****
161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is ***native badminton-first***.

165 **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not
166 the individual.

167

168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**
171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.

173

174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
175 Korea) + Denmark**
176 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
177 school-club & workplace systems), mature club infra, fixed-court recording.
178 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia
179 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
182 motion**, then expand to **pickleball-US** (out-build SwingVision) and
183 **badminton-Asia/Denmark** (density).

184

185 ---

186

187 ## 6. Risks & open questions (what would invalidate the thesis)

188

189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.

192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.

194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).

198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200 INR licensing.

201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).

204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205 academy self-serve before institutional deals.

206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.

208

209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

212

213 ---

214

215 ## 7. Sources

216

217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
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220 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
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224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)
225 market (2024);
226 <https://www.precedenceresearch.com/action-camera-market> (2024);
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are-there-in-the-world/ ;
228 <https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/> ;
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231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
232 - Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
233 <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
234 - Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
235 <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
241 - Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing>
;
242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
243 <https://www.badproplusapp.com/>
244

7. user (2026-06-12T14:42:37.134Z)

34-| | **? PENDING (needs action / not started)** | | | |
35-| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC** (`libx264` in `backend\pipeline\stitcher\compiler.py`) ? encode-side AVC patent exposure remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4\L40)** for HW encode. See §4. Decision: provision Ada-class GPU? |
36-| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic `setup.py` shadows the build system. | Design in `DEFERRED\_HARDENING\_PLAN.md` (#7) ? **awaiting your approval** before code. |
37-| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM\_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED\_HARDENING\_PLAN.md` (#16). |
38-| | **? IN-FLIGHT (in progress / watch)** | | | |
39-| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |
40-| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF\_MARKET\_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |
41-| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |
42-[Omitted long context line]
43-[Omitted long context line]
44-| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |
45-| C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5.
|

46-| C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL\JSON\CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-
player/match stats, clip gallery. See §6.
198-
199----
200-
201-## Changelog
202-
203-- **2026-06-07** ? Added the market-research-of-record (`PMF\_MARKET\_RESEARCH.md`):
204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205- "beachhead validation" workstream. Wired the summary into §6.
206-- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
207- audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
208- removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209- the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
210- `DEFERRED\_HARDENING\_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer

8. user (2026-06-12T14:42:38.609Z)

14:## Where we are (one paragraph)
23:## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
38:## Prioritized next steps
56:5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two
added questions:
57: "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the
warmest 2-3 academies
59: `output/MahadevpuraSingles\_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
60: physical-visit approach):** `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (recruit hand-out:
script, ground rules,
61: Phase-2 consent; owner fills ?/contact placeholders) + `docs/FIELD\_VISIT\_ANSWER\_SHEET.md`
(per-visit capture
62: form) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R scoring anchors + **pre-registered decision
rules** ? PROCEED /
63: PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
68:## Pointers / artifacts

9. user (2026-06-12T14:43:38.371Z)

Structured output provided successfully

10. assistant (2026-06-12T14:43:45.992Z)

Verification complete. The claim is real but overrated: the H6 ordering-capture gap is genuine
(and the Q5 "what number did THEY say first" field proves the kit's designers knew this
mattered), but H6 feeds none of the pre-registered decision triggers and the take-lower tiebreak
resolves the ambiguity conservatively, so severity is low.